# My white board writing from week 8

1) Based on the theoretical discussion presented at the page extract (pages 232–233) from the SAS Help for the Discriminant analysis procedure, I explained the SAS output related to the option pool = test in the discr2.sas code. For the IRIS data, we were testing the null hypotesis $H_0^{(2)}: \Sigma_1 = \Sigma_2 = \Sigma_3$ for the $k=3$ classes of iris plants based on the 4-dim. measurements ($p=4$). The test (denoted as $H_0^{(2)}$ on p. 232) is presented on p. 233 and it is a likelihood-ratio test that boils down to compare $-2\ln\lambda_2$, ($\lambda_2$ given in (5.2)) with a threshold related to the null distribution of the $-2\ln\lambda_2$ statistic (or equivalently, calculating the $p$-value for this statistic).

Under $H_0^{(2)}$, $-2\ln\lambda_2$ is (asymptotically) distributed as $\chi^2$ with $df = \frac{p(p+1)(k-1)}{2}$. In our case this is $\frac{4*5*(3-1)}{2} = 20$ df. The output gives $-2\ln\lambda_2 = 140.943$, and a $p$-value $< 0.001$. Hence we reject $H_0^{(2)}$ and decide to perform quadratic, instead of linear, discriminant analysis for the iris data.

2) I then started lecture 8 (Principal component analysis). The lecture notes are thorough and I abstain from reproducing them here.
I will only discuss an example about the effect of standardization on the principal component extraction. In cases of non-standardized units of measurement, this effect can be quite significant. The example is from the textbook (Johnston and Wichern, VI Ed., p. 437)
Let $p=2$ and we want to analyse the structure and the effect of the first principal component extracted from the 2-dim. vector. Assume that

$\Sigma = \begin{pmatrix} 1 & 4 \\ 4 & 100 \end{pmatrix}$ (i.e., variances are very heterogeneous).

This leads to $\lambda_1 = 100.16$, $\lambda_2 = .84$ as eigenvalues and

$e_1 = \begin{pmatrix} 0.04 \\ .999 \end{pmatrix}$, $e_2 = \begin{pmatrix} .999 \\ -0.04 \end{pmatrix}$ as respective normed eigenvectors.

Then the two principal components are:

$Y_1 = 0.04X_1 + .999X_2$ ; $Y_2 = .999X_1 - .04X_2$ .

When using only $Y_1$ we can explain $\frac{\lambda_1}{\lambda_1+\lambda_2} = .992 = 99.2\%$ of the variability. We also see that this first principal component is almost totally dominated by $X_2$ (the variable with the higher variance).

NOW let us standardize the measurements:

$Z_1 = \frac{X_1 - \mu_1}{1}$ ; $Z_2 = \frac{X_2 - \mu_2}{10}$ . The resulting correlation matrix (which is also the covariance matrix of $Z$'s) is:

$\rho = \begin{pmatrix} 1 & .4 \\ .4 & 1 \end{pmatrix}$, with eigenvalues $\lambda_1 = 1+.4 = 1.4$ ; $\lambda_2 = 1-.4 = .6$ and respective eigenvectors $e_1 = \begin{pmatrix} .707 \\ .707 \end{pmatrix}$, $e_2 = \begin{pmatrix} .707 \\ -.707 \end{pmatrix}$ and principal components

$\tilde{Y}_1 = .707\left(\frac{X_1 - \mu_1}{1}\right) + .707\left(\frac{X_2 - \mu_2}{10}\right) = .707(X_1 - \mu_1) + .0707(X_2 - \mu_2)$

$\tilde{Y}_2 = .707(X_1 - \mu_1) - 0.0707(X_2 - \mu_2)$

Now the first principal component is in fact more heavily dominated by $X_1$ and explains $\frac{1.4}{2} = 70\%$ only of the variability